

SE3082 – Parallel Computing - Lab08

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Exercise 01

```
[ec2-user@ip-172-31-10-115 ~]$ mpirun --oversubscribe -np 2 ./hello
Hello World! From rank 0 machine ip-172-31-10-115.ec2.internal
Just a normal process From rank 1 machine ip-172-31-10-115.ec2.internal
[ec2-user@ip-172-31-10-115 ~]$
```

```
Just a normal process From rank 3 machine ip-172-31-10-115.ec2.internal
Just a normal process From rank 1 machine ip-172-31-10-115.ec2.internal
Just a normal process From rank 2 machine ip-172-31-10-115.ec2.internal
Hello World! From rank 0 machine ip-172-31-10-115.ec2.internal
[ec2-user@ip-172-31-10-115 ~]$
```

```
[ec2-user@ip-172-31-10-115 ~]$ mpirun --oversubscribe -np 4 ./message1
Normal process: rank 0
Normal process: rank 2
Sending message from rank 1 to rank 3
In rank 3, y = 30
```

Exercise 02

```
[ec2-user@ip-172-31-10-115 ~]$ nano sum_openmp.c
[ec2-user@ip-172-31-10-115 ~]$ gcc -O2 -fopenmp -o sum_openmp sum_openmp.c
[ec2-user@ip-172-31-10-115 ~]$ export OMP_NUM_THREADS=1
[ec2-user@ip-172-31-10-115 ~]$ ./sum_openmp
Number of threads: 1
Total sum = 50000005000000
Expected sum = 50000005000000
Execution time = 0.003280 seconds
[ec2-user@ip-172-31-10-115 ~]$ export OMP_NUM_THREADS=2
[ec2-user@ip-172-31-10-115 ~]$ ./sum_openmp
Number of threads: 2
Total sum = 50000005000000
Expected sum = 50000005000000
Execution time = 0.004788 seconds
[ec2-user@ip-172-31-10-115 ~]$ export OMP_NUM_THREADS=4
[ec2-user@ip-172-31-10-115 ~]$ ./sum_openmp
Number of threads: 4
Total sum = 50000005000000
Expected sum = 50000005000000
Execution time = 0.003501 seconds
[ec2-user@ip-172-31-10-115 ~]$ export OMP_NUM_THREADS=8
[ec2-user@ip-172-31-10-115 ~]$ ./sum_openmp
Number of threads: 8
Total sum = 50000005000000
Expected sum = 50000005000000
Execution time = 0.003627 seconds
[ec2-user@ip-172-31-10-115 ~]$ export OMP_NUM_THREADS=12
[ec2-user@ip-172-31-10-115 ~]$ ./sum_openmp
Number of threads: 12
Total sum = 50000005000000
Expected sum = 50000005000000
Execution time = 0.003833 seconds
[ec2-user@ip-172-31-10-115 ~]$
```

Exercise 03

```
[ec2-user@ip-172-31-10-115 ~]$ nano pi_openmp.c
[ec2-user@ip-172-31-10-115 ~]$ gcc -O2 -fopenmp -o pi_openmp pi_openmp.c
[ec2-user@ip-172-31-10-115 ~]$ export OMP_NUM_THREADS=1
[ec2-user@ip-172-31-10-115 ~]$ ./pi_openmp
Number of threads: 1
Total points = 10000000
Points inside circle = 7853466
Estimated Pi = 3.1413864000
Execution time = 0.099050 seconds
[ec2-user@ip-172-31-10-115 ~]$ export OMP_NUM_THREADS=2
[ec2-user@ip-172-31-10-115 ~]$ ./pi_openmp
Number of threads: 2
Total points = 10000000
Points inside circle = 7853163
Estimated Pi = 3.1412652000
Execution time = 0.066527 seconds
[ec2-user@ip-172-31-10-115 ~]$ export OMP_NUM_THREADS=4
[ec2-user@ip-172-31-10-115 ~]$ ./pi_openmp
Number of threads: 4
Total points = 10000000
Points inside circle = 7852191
Estimated Pi = 3.1408764000
Execution time = 0.137175 seconds
[ec2-user@ip-172-31-10-115 ~]$ export OMP_NUM_THREADS=8
[ec2-user@ip-172-31-10-115 ~]$ ./pi_openmp
Number of threads: 8
Total points = 10000000
Points inside circle = 7852810
Estimated Pi = 3.1411240000
Execution time = 0.067678 seconds
[ec2-user@ip-172-31-10-115 ~]$ export OMP_NUM_THREADS=12
[ec2-user@ip-172-31-10-115 ~]$ ./pi_openmp
Number of threads: 12
Total points = 10000000
Points inside circle = 7853327
Estimated Pi = 3.1413308000
```

Exercise 04

For **Exercise 2**, record:

Processes	Time (seconds)
1	0.003280
2	0.004788
4	0.003501
8	0.003627
12	0.003833

For **Exercise 3**, record:

Processes	Time (seconds)
1	0.099050
2	0.066527
4	0.137175
8	0.067678
12	0.068967

Speedup = T_1/T_N

From exercise 02

Processes	Time (seconds)	Speedup
1	0.003280	1.0000
2	0.004788	0.6850
4	0.003501	0.9369
8	0.003627	0.9043
12	0.003833	0.8557

From exercise 03

Processes	Time (seconds)	Speedup
1	0.099050	1.0000
2	0.066527	1.4889
4	0.137175	0.7221
8	0.067678	1.4635
12	0.068967	1.4362

Exercise 05

```
[ec2-user@ip-172-31-10-115 ~]$ mpirun --oversubscribe -np 4 ./exercise5
Rank 0: Normal process
Rank 1: Sending message to rank 2
Rank 2: Normal process
Rank 3: Waiting for message from rank 1
```

```
[ec2-user@ip-172-31-10-115 ~]$ mpicc -o message2_bsend message2_bsend.c
[ec2-user@ip-172-31-10-115 ~]$ mpirun --oversubscribe -np 4 ./message2_bsend
Rank 2: Normal process
Rank 1: Sending array to rank 3 using MPI_Bsend
Rank 1: Buffered send completed
Rank 0: Normal process
Rank 3: Received values:
0 10 20 30 40 50 60 70 80 90
```

Exercise 06

```
[ec2-user@ip-172-31-10-115 ~]$ nano exercise6.c
[ec2-user@ip-172-31-10-115 ~]$ mpicc -o exercise6 exercise6.c
[ec2-user@ip-172-31-10-115 ~]$ mpirun --oversubscribe -np 4 ./exercise6
Rank 2: Sending value 50 to rank 3
Rank 1: Sending value 30 to rank 3
Rank 3: Received 50 from rank 2
Rank 0: Normal process
```

Exercise 07

```
[ec2-user@ip-172-31-10-115 ~]$ nano exercise7.c
[ec2-user@ip-172-31-10-115 ~]$ mpicc -o exercise7 exercise7.c
[ec2-user@ip-172-31-10-115 ~]$ mpirun --oversubscribe -np 4 ./exercise7
Rank 2: Sending 50 using MPI_Bsend
Rank 0: Normal process
Rank 1: Sending 30 using MPI_Bsend
Rank 3: Received 30 from rank 1
[ec2-user@ip-172-31-10-115 ~]$
```